CHAPTER

# 13 Surgical audit and research

## Learning objectives

To understand:

- The planning and conduct of surgical audit and research
- How to write up a project
- How to review a journal article and determine its value

## INTRODUCTION

Surgeons are innovators and a key aspect of a surgical career is to constantly adapt, tweak and improve surgical techniques and treatments to provide the best outcomes for those under our care. In addition, few others in the hospital use more technology or devices on a day-to-day basis than surgeons. It is therefore beholden upon all surgeons to be able to critically evaluate both our individual performance and the impact of adaptations in techniques, devices and treatment pathways on the individual and collective outcomes of our patients.

Involvement in research and audit activities will form a stable cornerstone within a long and successful surgical career. The aim of this chapter is to outline both how to undertake a successful audit cycle and how to design and conduct a surgical research study. Key aspects associated with enhanced chances of a successful project are discussed, including how collaboration with others can be crucial.

Large numbers of clinical papers appear in the surgical literature every year. Many are flawed, and it is important that a surgeon has the skills to examine publications critically. The best way to develop a critical understanding of the research and audit undertaken by others is to perform studies of one's own. The hardest part of audit and research is writing it up, and the hardest article to write is the first. This chapter contains the information required to write a surgical paper and to evaluate the publications of others.

## AUDIT OR RESEARCH?

Health professionals are expected to undertake audit and service evaluation as part of quality assurance. These usually involve minimal additional risk, burden or intrusion for participants. It is important to determine at an early stage whether a project is audit or research, and sometimes that is not as easy as it seems. The decision will determine the framework in which the study is undertaken. In the UK, the Health Research Authority (HRA) has developed a decision tool to help decide whether your project is classified as research (http://www.hra-decisiontools.org.uk/research/). This tool crystallises the differentiation between audit and research to three overarching questions:

1. Are the participants in your study randomised to different groups?
2. Does your study protocol demand changing treatment/care/services from accepted standards for any of the patients/service users involved?
3. Is your study designed to produce generalisable or transferable findings?

Although the first two questions are simple to comprehend, the third can create some confusion at times. The HRA states that, in this context, 'generalisable' means the findings can be reliably extrapolated from the study to a broader population of patients/service users and/or applied to settings or contexts other than those in which they were tested. The majority of audits can be assumed to be hypothesis generating as they would require subsequent prospective testing in a new population before findings could be considered as new 'evidence' – as such they do not fulfil this generalisability criterion. Finally, in this context, 'transferable' means that the findings of a qualitative study can be assumed to be applicable to a similar context or setting. Most qualitative studies are not usually generalisable but can quite often be considered to be transferable.

Further useful information on classifying your proposed project can be found in the HRA leaflet 'Differentiating clinical audit, service evaluation, research and usual practice/surveillance work in public health' (http://www.hra-decisiontools.org.uk/research/docs/DefiningResearchTable_Oct2017-1.pdf).

## AUDIT AND SERVICE EVALUATION

Clinical audit is a process used by clinicians who seek to improve patient care. The process involves comparing aspects of care (structure, process and outcome) against explicit criteria and defined standards. Keeping track of personal

outcome data and contributing to a clinical database ensures that a surgeon's own performance is monitored continuously and can be compared with a national data set to ensure compliance with agreed standards. Involvement in active audit processes is also an essential component of revalidation for the individual surgeon in the UK. If care falls short of the guidance standard being compared against, some change in the way that care is organised should be proposed. This change may be required at one of many levels. It might be an individual who needs training or surgical equipment that needs replacing. At times, the change may need to take place at the team level. Sometimes, the only appropriate action is change at an institutional level (e.g. a new antibiotic policy), regional level (provision of a tertiary referral centre) or, indeed, national level (screening programmes and health education campaigns).

There are two main types of audit in common practice – single site/local audits and multisite regional, national or international audits. Both are designed to improve the quality of care. In an ideal world local audits might identify needs closest to the patient, which can then be further investigated in multisite larger scale audits. For example, hospital topics are often identified at departmental morbidity and mortality meetings, where issues relating to patient care are discussed. The reporting process might identify a possible national issue, and a national or international audit could be designed to be delivered by local surgical teams.

Audits are formal processes that require a structure. The following steps are essential to establish an audit cycle:

1. Define the audit question in a multidisciplinary team.
2. Identify the body of evidence and current standards.
3. Design the audit to measure performance against agreed standards based on strong evidence. Seek appropriate advice (local audit department in the UK) and ensure institutions have agreed to undertake the audit.
4. Measure over an agreed interval.
5. Analyse results and compare performance against agreed standards.
6. Undertake gap analysis:
   - **a** if all standards are reached, reaudit after an agreed interval;
   - **b** if there is a need for improvement, identify possible interventions such as training, and agree with the involved parties.
7. Reaudit.

A new type of audit that has developed significant traction in surgery over recent years is the 'multicentre snapshot audit', whereby many collaborators across multiple hospitals prospectively collate anonymised patient-level data for a specific condition, presentation or intervention over a short time period of normally around 6–8 weeks. This allows exploration of differences in patients, techniques and management across the cohort to identify areas of practice variability that may result in apparent differences in outcome. These studies generate hypotheses and identify areas where further prospective research is needed. Key advantages of these snapshot audits are their easy accessibility and the fact that they can be conducted at almost zero cost, so they can be an excellent means of bringing a new group together to collaborate and create contemporaneous and 'real-world' data together.

## IDENTIFYING A RESEARCH TOPIC

Research is designed to generate new knowledge and might involve testing a new treatment or regimen. Once an idea has been formed, or a question asked, it needs to be transformed into a hypothesis. It is helpful to approach surgeons who regularly publish articles and who have a special interest in the subject area being considered. As ideas are suggested, it is important to consider whether the question posed really matters. Spending time refining the question (hypothesis) is probably the most important part of the process. Choosing the wrong topic can lead to many wasted hours. Once a topic has been identified, it is also important not rush into the study. The worst possible outcome is to find at the end of a long arduous study that the research has already been performed or that the chosen methodology did not support investigation of the primary/secondary outcomes.

The first port of call for information is the Internet (with assistance as needed from a medical librarian). Current articles about the proposed research should be retrieved; review articles and meta-analyses can be particularly helpful. It is very important to learn how to do an accurate and efficient search as early as possible. Collections of reviews are available – the Cochrane Collaboration brings together evidence-based medical information and is available in most libraries. Once information on the subject has been obtained and the relevant literature identified, it is important that these are carefully perused. It is not sufficient to just read the abstract! Further information is given in *Table 13.1*.

An excellent source of ideas where research is needed can come from reviewing high-quality national guidelines such as those produced by the UK National Institute for Health and Care Excellence (NICE) on a particular area of interest, many of which include a section beneath the headline guidance being made on 'recommendations for research'. This section is populated after the currently available evidence for an intervention or treatment has been reviewed by the expert team and found lacking. Designing a research project to cover one or more of these agreed areas can be easily justified to both funders and clinicians alike.

Finally, there is an increased number of Priority Setting Partnerships across all aspects of surgery, including those formally undertaken by the James Lind Alliance and by others run by surgical associations and their patient–partner groups. These partnerships consist of patients, carers, healthcare professionals and organisations or charities representing people with the particular condition. They focus on identifying and prioritising research gaps or important specific questions for

**Archibald Leman Cochrane**, 1909–1988, Director of the UK Medical Research Council Epidemiology Unit, Cardiff, UK, after whom the Cochrane Collaboration is named.

**TABLE 13.1** Electronic information sites.

| Database | Producer | Coverage | Availability |
|---|---|---|---|
| PubMed http://www.ncbi.nlm.nih.gov/pubmed | US National Library of Medicine (NLM) | PubMed comprises more than 25 million citations for biomedical literature from MEDLINE, life science journals and online books<br>Citations may include links to full-text content from PubMed Central and publisher websites | Internet |
| PubMed Central http://www.ncbi.nlm.nih.gov/pmc | US National Institutes of Health (NIH) free digital archive | Full-text archive of biomedical and life sciences journal literature at the US National Institutes of Health's National Library of Medicine | Internet |
| EMBASE http://embase.com | EMBASE | Providing extensive coverage of peer-reviewed biomedical literature, along with indexing, searching and information management tools | Subscription |
| CINAHL https://www.ebsco.com/products/research-databases/cinahl-database | CINAHL is owned and operated by EBSCO Publishing | Cumulated index to nursing and allied health literature | Subscription |
| Cochrane Collaboration and Library http://uk.cochrane.org | Global independent network of researchers, professionals, patients, carers and people interested in health to produce credible, accessible health information that is free from commercial sponsorship and other conflicts of interest | Preparing, updating and promoting the accessibility of Cochrane Reviews published online in The Cochrane Library | Internet |

which additional new research is needed to answer them. Again, creating research projects in these areas is likely to be well received; sometimes such studies are also prioritised for funding support.

It is also helpful to seek support from specific networks set up to support health research. In the UK, the National Institute for Health Research (NIHR) runs the Research Design Service (RDS), which provides free and confidential advice on research design, writing funding applications and obtaining public engagement in research for all researchers. There are also a number of training courses available in research methodology and application.

## FORMING A TEAM

One of the most common reasons for the failure of an otherwise good research project is failure to involve others. Only the smallest single-centre project can be delivered by an individual researcher working alone; almost any project worth doing will need a team to deliver it. This team can bring the necessary skills and experience to help bring the project to fruition but also, and perhaps more importantly, it can provide the momentum required to keep pushing a project through to completion when the inevitable hurdles are met.

There may be local colleagues who form a natural team for a project, perhaps with the oversight of an experienced trainer or mentor. Another solution can be to get involved in a collaborative research group. Surgery has led the way with collaborative research working over recent years.

The first trainee-level research collaborative in the UK was formed in 2008 when a group of surgical trainees who shared the same frustrations around the challenges of conducting high-quality research while engaged in a full-time training programme came together to create the West Midlands Research Collaborative (WMRC). The premise was simple: to create and conduct prospective research projects that simultaneously collate data from across all of the members' units and to take advantage of the rotation of trainees' postings between units to ensure project longevity and thus enable longer term outcome collection. By achieving a critical mass of engaged members in these projects, the collective momentum ensured completion even if individuals were unable to personally contribute in a consistent manner because of examinations, family life or busy clinical periods. Such research collaboratives can be most effective in undertaking two key types of study: (i) simple randomised controlled trials (RCTs) and (ii) multicentre snapshot audits (see ***Audit and service evaluation***).

The first RCT undertaken by the WMRC was the ROSSINI trial, which explored the clinical effectiveness of a simple wound-edge protection device in reducing wound infections after abdominal surgery. A network of trainees mobilised 21 units for the trial and together they completed the trial 2 months ahead of schedule, having randomised 760 patients over a 23-month period, completing in January 2013. This achievement galvanised the research collaborative model and stimulated other new groups to form.

There are now general surgical research collaboratives in every region of the UK and national collaboratives for each surgical subspeciality area such as neurosurgery and cardiothoracic surgery. Many other countries with rotational surgical training programmes have also formed their own parallel collaboratives, including Australia, Portugal, Italy, The Netherlands and Canada. The collaborative movement has

established a new paradigm for evidence-based surgical practice, engaged thousands of surgical trainees and their consultant mentors and created an active network of research active clinicians at many hospitals across the world.

In the UK, trainee collaboratives have, to date, developed at least 10 RCTs and been awarded competitive grant funding worth over £8 million. The model has also extended to medical student collaboratives (STARSurg), and all 42 medical schools in the UK now have an active network student research collaborative. More recently, similar research collaboratives have also formed, utilising the established core principles, in non-surgical specialities such as anaesthetics, gastroenterology and elderly care.

All of these collaborative groups work on a principle of complete inclusivity – any interested person is very welcome to get involved in the collaborative; both in existing projects and in suggesting new ideas. People can join at any stage from medical student to consultant. Anyone interested in surgical research should seek out their local or national surgical research collaborative group and get involved.

## PROJECT DESIGN

During the first phase, it is important to keep in the mind some important questions (*Summary box 13.1*).

**Summary box 13.1**

Questions to answer before undertaking research

- Why do the study?
- Will it answer a useful question?
- Is it practical?
- Can it be accomplished in the available time and with the available resources?
- Will the project benefit from collaboration to increase numbers or make best use of high-technology equipment?
- What findings are expected?
- What are the research governance requirements?
- What are the ethical issues?
- What impact could it have?

There are many different types of scientific study. The design used depends on the study. Time spent carefully designing a potential project is never wasted. An RCT is regarded as one of the best methods of scientific research; however, much surgical practice has been advanced through other different types of study such as those listed in *Table 13.2*. For example, testing a new type of operation often requires a pilot study to assess feasibility, which is then followed by a formal RCT. The introduction of innovative surgical techniques may require novel handling, and recommendations have been made by the IDEAL collaborators (see *Further reading*).

Research can be qualitative or quantitative. Quantitative research allows hard facts to speak for themselves. A medical condition is analysed systematically using hard, objective end points such as death or major complications, which should be clearly defined. For example, surgical complications are now classified using the Clavien–Dindo system. In qualitative research, data often come from patient narratives, and the psychosocial impact of the disease and its treatment are analysed; for example, narratives from patients with breast cancer. These kinds of data are often collected using quality-of-life measurements. A variety of different quality-of-life questionnaires exist to suit several different clinical situations. Much of the best research is both quantitative and qualitative. Recently, the importance of outcomes from the patient's perspective has been emphasised: patient-reported outcome measures (PROMs) are now an important component of the evaluation of surgical procedures.

**TABLE 13.2** Types of study.

| Type of study | Definition |
|---|---|
| Observational | Evaluation of condition or treatment in a defined population |
| | Retrospective: analysis of past events |
| | Prospective: contemporaneous collection of data |
| Case–control | Series of patients with a particular disease or condition compared with matched control patients |
| Cross-sectional | Measurements made on a single occasion, not looking at the whole population but selecting a small similar group and expanding results |
| Longitudinal | Measurements taken over a period of time, not looking at the whole population but selecting a small similar group and expanding results |
| Experimental | Two or more treatments are compared. Allocation to treatment groups is under the control of the researcher |
| Randomised | Two or more randomly allocated treatments |
| Randomised controlled | Includes a control group with standard treatment |

Research should be focused according to institutional, national and international strategies. As finances for health care are always limited, it is important to consider including a cost–benefit analysis in any major area of research so that the value of the proposed intervention or change in treatment can be assessed. The NIHR provides the framework through which the Department of Health maintains and manages the research, research staff and research infrastructure of the National Health Service (NHS) in England.

### Sample size

Calculating the number of patients required to perform a satisfactory investigation is an important prerequisite to any study. An incorrect sample size is probably the most frequent reason for research being invalid. Often, surgical trials are marred by the possibility of error caused by the inadequate number of patients investigated.

**Pierre-Alain Clavien**, contemporary, Professor of Surgery, Zurich, Switzerland.
**Daniel Dindo**, contemporary, surgeon, Zurich, Switzerland.

- **Type I error**. Benefit is perceived when really there is none (false positive).
- **Type II error**. Benefit is missed when it was there to be found (false negative).

Calculating the number of patients required in the study can overcome this bias. Unfortunately, it often reveals that a larger number of patients is needed for the study than can possibly be obtained from available local resources. This usually means expanding enrolment by running a multicentre study – which has the added benefit of improving the external validity of findings. More patients will need to be randomised than the final sample size to take into account patients who die, drop out or are lost to follow-up; this is known as the attrition rate. A longer time from trial entry to primary outcome assessment will result in an increased attrition rate of participants.

The following is an example calculation for a study to recruit patients into two groups. In order to calculate a sample size, it is now common practice to set the level of power for the study at 90% with a 5% significance level. This means that, if there is a difference between study groups, there is a 90% chance of detecting it. Based on previous studies, realistic expectations of differences between groups (i.e. the magnitude of the effect seen from utilising the intervention under study), according to the best available evidence, should be used to calculate the sample size. The formula below uses the results of a reduction in event rate from 30% to 20% (e.g. a new treatment expected to reduce the complication rate such as wound infection from 30% = $r$ to 20% = $s$).

$$9 \times \frac{[r(100 - r) + s(100 - s)]}{(r - s)^2}$$

$$\text{e.g. } 9 \times \frac{[30(100 - 30) + 20(100 - 20)]}{(30 - 20)^2}$$

$$= 333 \text{ needed in each group}$$

## Eliminating bias

It is important to imagine how a study could be invalidated by thinking of things that could go wrong. One way to eliminate any bias inherent in the data collection is to have observers or recorders who do not know which treatment has been used (blinded observer). It might also be possible to ensure that the patient is unaware of the treatment allocation (single blind). In the best randomised studies, neither patient nor researcher is aware of which therapy has been used until after the study has finished (double blind). Randomised trials are essential for testing new drugs. In practice, however, in some surgical trials, randomisation may not be possible or ethical.

## Study protocol

Now that the research question has been decided, and it has been checked that sufficient patients should be available to enrol into the study, it is time to prepare the detail of the trial. At this stage, a study protocol should be constructed to define the research plan. It should contain the background of the proposed study, the aims and objectives, a clear methodology, definitions of population and sample sizes and methods of proposed analysis. It should include the patient numbers, inclusion and exclusion criteria and the timescale for the work. The protocol should be detailed enough for another party to come along in the future and theoretically replicate the study. It is useful to construct a flow diagram giving a clear summary of the research protocol and its requirements (*Figure 13.1*). It is helpful to imagine the paper that will be written about the study before the study is performed. This may prevent errors in data collection.

When a study is planned, sufficient time should be reserved at the beginning for fund-raising and obtaining ethical, regulatory and or other approvals (e.g. HRA). Time for data analysis and preparation of publication needs to be included in funding applications. The cost of any non-routine investigations and extra treatments should be identified and covered by the research grant in line with national guidance (in the UK, the Attributing the costs of health and social care Research and Development [AcoRD] guidance; https://www.gov.uk/government/publications/guidance-on-attributing-the-costs-of-health-and-social-care-research).

A data collection form should be designed or a computer collection package developed. If data are collected on a computer, appropriate safeguards for privacy, confidentiality and data quality will be necessary to comply with legislation. At this stage it is important to consider any validation requirements and needs for open access, either in a recognised archive (e.g. the UK Data Archive) or in an institutional repository. Any form of data collection needs to be quality assured. The quality assurance process will include training, standard operating procedures as well as monitoring and checking a certain sample of the data. At the end of data collection and analysis, a final database with all data should be locked and kept for future reference in a safe location. A data-archiving policy with a nominated data custodian should be in place.

Research is no longer confined by institutional or even geographical boundaries. Collaborative research groups in surgery at a national or international level have come together to undertake high-quality surgical research in recent years, aided by online communication and the availability of secure electronic databases such as REDCap™. The SUNRRISE trial shown in *Figure 13.1* was undertaken by researchers from two trainee-led research collaborative groups across the UK, in conjunction with a parallel collaborative group in Australia. All patients were included within the same study cohort in real time: the Australian sites were effectively identical to those in the UK because of online electronic randomisation systems and online live data capture via REDCap.

Some publishers require registration of a study at the time it is set up on a publicly available database (e.g. the World Health Organization's recognised registries such as ISRCTN, EudraCT and ClinicalTrials.gov). It is becoming increasingly popular to consider publication of a protocol paper.

## Regulatory framework

In the UK, the implementation of the UK Policy Framework for Health and Social Care Research provides a framework

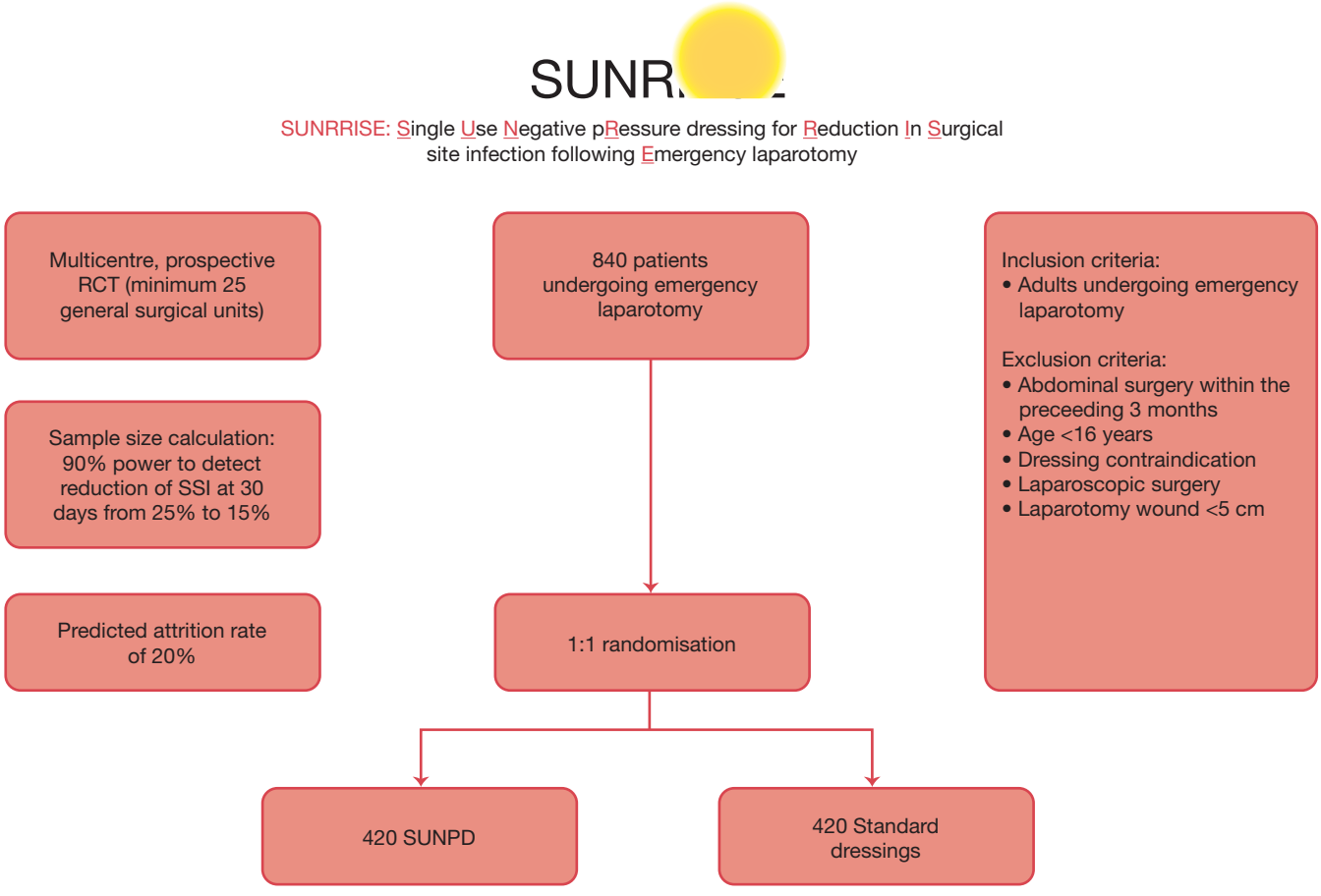


**Figure 13.1** SUNRRISE trial: protocol summary and recruitment flowchart. Permission is granted by Birmingham Clinical Trials Unit, University of Birmingham, UK. RCT, randomised controlled trial; SSI, surgical site infection; SUNPD: single-use negative-pressure dressing.

that enhances the integrity of the study and includes requirements for sponsorship by an institution to ensure the following: peer review, independent ethics review, compliance with data protection principles, financial probity, dissemination and management of intellectual property.

Sponsorship is defined by the HRA as the individual, company, institution or organisation that takes on ultimate responsibility for the initiation, management (or arranging the initiation and management) of and/or financing (or arranging the financing) for that research. The sponsor takes primary responsibility for ensuring that the design of the study meets appropriate standards and that arrangements are in place to ensure appropriate conduct and reporting (https://www.hra.nhs.uk/planning-and-improving-research/research-planning/roles-and-responsibilities/#sponsor).

## Peer review

Once the protocol is finalised, formal peer review is needed. In the UK, evidence of peer review will be needed before submitting an application to a research ethics committee and for HRA approval. Many funders of research will undertake their own independent peer review. There is usually feedback from this process that can provide valuable advice about the study.

## Ethics

In the first instance, common sense is the best guide to whether or not a study is ethical. It is still important to seek advice from an independent research ethics committee whenever research is contemplated.

In the UK the requirement is that an NHS Research Ethics Committee (NHS REC) provides an independent ethical review of all health and social care research if it involves patients and/or carers. The Governance arrangements for Research Ethics Committees (GafREC) provides detailed guidance about NHS REC review requirements. The application for NHS REC review is made using the Integrated Research Application System (IRAS). IRAS enables entry of information about the project once, instead of duplicating information in separate application forms for regulators.

If the study does not require review by an NHS REC, the need for an independent ethical review should still be considered. Universities have developed their own ethical review infrastructure and this will be institute specific and location specific. For collaborative research, local ethical review should be obtained where possible, and developing a local ethics infrastructure should be considered if it does not already exist. Duplication of ethical review should be avoided.

Ethics committee forms may seem long and detailed, but it is important that these are filled in correctly as this helps to

prepare the investigators for all practical aspects of the project. All dealings with ethics committees should be intelligent and courteous. It is important to attend the meeting at which the study will be discussed, if invited, as it provides a forum for direct communication in relation to the study. It can save time as possible concerns of the ethics committee can be addressed at the time, avoiding lengthy correspondence.

## Regulatory approvals

Interventional clinical or device trials are regulated by the Medicines and Healthcare products Regulatory Agency (MHRA) in the UK. Researchers are encouraged to use an existing and established international register such as ISRCTN or ClinicalTrials.gov to ensure that the public is aware of a trial before recruitment of the first participant. Trials involving sites specifically in EU countries must be registered in the EU Clinical Trials Register. Trials should be registered before applying to the MHRA for a clinical trial authorisation via the MHRA submissions portal and researchers must ensure that the registry is kept up to date and trial results are uploaded in the appropriate timeframe. This can be a complicated and trying process, and support should be sought from the investigators' employing institution. Editors of the major surgical journals now agree that all clinical trials should have been registered before an article relating to a trial can be published.

All studies undertaken with NHS patients and/or carers will need HRA Approval and confirmation of capacity and capability from NHS sites. Studies involving animals require approval from statutory licensing authorities. In the UK this is the Home Office. Animal research should be based on ARRIVE guidelines (Animal Research: Reporting of In Vivo Experiments).

## Research integrity

In 2013, Universities UK, in collaboration with major funders of research, developed The Concordat to Support Research Integrity, which sets out key commitments to ensure a high standard in research. It highlights the principles and professional responsibilities of researchers and research institutions that are fundamental to the integrity of research wherever it is undertaken. These centre on:

- honesty in all aspects of research;
- accountability and transparency in the conduct of research;
- professional courtesy and fairness in working with others;
- good stewardship of research.

A study should not under any circumstances commence until appropriate approvals have been granted and compliance with the principles of research integrity is ensured. A helpful international summary of country-specific approval requirements is available from NIH (https://clinregs.niaid.nih.gov).

# STATISTICAL ANALYSIS

Both audit and research commonly require statistical analysis. Many surgeons find the statistical analysis of a project the most difficult part. It is also the most commonly criticised part of papers written by clinicians. There are many useful books about statistics (see ***Further reading***); if in any doubt, a statistician will be pleased to give assistance. Statisticians should be consulted before research or audit has been conducted rather than being presented with the data at the end; they often give helpful advice over study design and can be an important part of the project team.

The following terms are frequently used when summarising statistical data:

- **Mean**: the result of dividing the total by the number of observations (the average).
- **Median**: the middle value with equal numbers of observations above and below – used for numerical or ranked data.
- **Mode**: the value with the highest frequency observed – used for nominal data collection.
- **Range**: the largest to the smallest value.

The most important decision for analysis is whether the distribution of the data is normal (i.e. parametric or non-parametric). Normally, distributed data have a symmetrical bell-shaped curve, and the mean, median and mode all lie at the same value. The type of data collected determines which statistical test should be used.

1. Numerical and normally distributed (e.g. blood pressure) – use an unpaired *t*-test to compare two groups or a paired *t*-test to assess whether a variable has changed between two time points.
2. Numerical but not normally distributed (e.g. tumour size) – use a Mann–Whitney *U*-test to compare two groups or a Wilcoxon signed rank test to assess whether a variable has increased/stayed the same/decreased between two time points.
3. Categorical (e.g. admitted or not admitted to an intensive care unit) – a chi-squared test can be used to compare two groups.

(**Note**: the use of these and any other statistical tests may benefit from professional advice.)

Confidence intervals are the best guide to the possible range in which the true differences are likely to lie. A confidence interval that includes zero usually implies a lack of statistical significance.

Scientists usually employ probability (*P*-values) to describe statistical chance. A *P*-value <0.05 is commonly taken to imply a true difference. It is important not to forget that $P = 0.05$ simply means that there is only a 1:20 chance that the differences between the variables would have happened by chance when in fact there is no real difference. If enough variables are examined in any study, significant differences will occur simply

**Henry Berthold Mann**, 1905–2000, and his student **Donald Ransom Whitney**, 1915–2007, Ohio State University, OH, USA, published their seminal paper 'On a test of whether one of two random variables is stochastically larger than the other' in 1947.
**Frank Wilcoxon**, 1892–1965, born County Cork, Ireland, American chemist and statistician.

by chance. Trials with multiple end points or variables require more sophisticated analysis to determine the significance of individual risk factors. Univariable or multivariable logistic regression analysis techniques may be appropriate.

Statistics simply deal with the chance that observations between populations are different and should be treated with caution. Clinical results should show clear differences. If statistics are required to demonstrate differences between results, it is likely that they are unlikely to have major clinical significance.

### Computer software packages available

Statistical computer packages offer a quick way of analysing descriptive statistics such as mean, median and range, as well as the most commonly used statistical tests such as the chi-squared test. Various packages are available commercially and are useful tools in data analysis.

## ANALYSING A SCIENTIFIC ARTICLE

The simplest way to analyse an article from a scientific journal is to look at the checklist of requirements for good scientific research. A group of scientists and editors developed the Consolidated Standards of Reporting Trials (CONSORT) statement to improve the quality of reporting of RCTs. Looking in detail at the study design is often the best way of deciding whether a trial is of value. The CONSORT document includes a checklist for the conduct of good RCTs (*Table 13.3*). Often clinicians overlook biases that others find obvious to detect, which can have a profound influence on the outcome of any study. Even the randomised design does not always guarantee quality, and a core component of systematic review is the grading of trial quality; several scoring systems have been developed (e.g. Jadad score). Recent guidelines have been published formalising the methods of systematic review and meta-analysis (Preferred Reporting Items for Systematic Reviews and Meta-Analyses [PRISMA] guidelines), and also many other types of article. These can be found in the instructions to authors of most surgical journals, which will now only accept articles that follow those rules.

**TABLE 13.3** Consolidated Standards of Reporting Trials (CONSORT) checklist for authors.

| Heading | Subheading | Descriptor |
|---|---|---|
| Title | | Identify as randomised trial |
| Abstract | | Structured format |
| Introduction | | Prospectively defined hypotheses, clinical objective |
| Methods | Protocol | Study population |
| | | Intervention, timing |
| | | Primary and secondary outcome |
| | | Statistical rationale |
| | | Stopping rules |
| | Assignment | Unit of randomisation |
| | | Method: allocation schedule |
| | Masking (blinding) | |
| Results | Participant flow and follow-up | Trial profile, flow diagram |
| | Analysis | Estimated effect of intervention |
| | | Summary data with appropriate inferential statistics |
| | | Protocol deviation |
| Comment | | Specific interpretation of study |
| | | Sources of bias |
| | | External validity |
| | | General interpretation |

From the CONSORT statement: *Journal of the American Medical Association* 1996; 276: 637–9.

## PRESENTING AND PUBLISHING AN ARTICLE

There is no point in conducting a research or audit project and then leaving the results unreported. Even when results are negative, they are worth distributing; no project if properly conducted is worthless. Under-reporting of negative outcomes causes a systematic bias in the literature in favour of positive trials. Most studies do not provide dramatic results, and few surgeons publish seminal articles.

The key to both presentation and publication is to decide on the message and then aim for an appropriate forum. Big important randomised studies or national audits merit presentation at national or international meetings and publication in international journals. Small observational studies and audits are more often accepted for presentation at regional meetings and for publication in smaller specialist journals. Help and advice from clinicians familiar with presentation and publication are invaluable at this stage. The most important piece of advice is to follow accurately the instructions for journal submission. Most international meetings will accept presentations eagerly (especially by poster) as this increases the attendance at a conference.

Most surgeons publish research in peer-reviewed journals. The work that is submitted is checked anonymously by other surgeons before publication. If in doubt about whether to submit to a journal, many editors will give advice about the suitability of an article for submission to their journal. It is usually free to publish in surgical journals since the cost of refereeing and editing is borne by the journal subscriber. A second model of publication is becoming more prevalent: open access, in which the author pays. This ensures that all research is visible to anyone, by pushing the costs of the editorial process onto the study budget. It may well become standard in future.

Convention dictates that articles are submitted in IMRAD form: introduction, methods, results and discussion. Increas-

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ingly, electronic publication and the Internet is changing the face of scientific publication and, in the next decade, these restrictions on style may disappear. For now, the IMRAD format remains inviolable. The length of an article is important: a paper should be as long as the size of the message. Readers of large randomised multicentre trials need to know as much detail about the study as possible; reports on small and simple trials should be brief.

- **Introduction**. This should always be short. A brief background of the study should be presented and then the aims of the trial or audit outlined.
- **Methods**. The methodology and study design should be given in detail. It is important to identify potential biases. New techniques or investigations should be detailed in full; if they are common practice or have been described elsewhere, this should be referenced instead of described.
- **Results**. Results are almost always best shown diagrammatically using tables and figures. Results shown in the form of a diagram need not then be duplicated in the text.
- **Discussion**. It is important not to repeat the introduction or reiterate the results in this section. The study should be interpreted intelligently and any suggestions for future studies or changes in management should be made. It is prudent not to indulge in flights of fantasy or wild imagination about future possibilities; most journal editors will delete these. Recently, a standard format for the discussion section has been promoted, and journals such as the *BMJ* are keen that authors use it.
- **References**. This section should include all relevant papers recording previous studies on the subject in question. The reference section does not usually have to be exhaustive, but should include up-to-date articles. Remember to present the references in the style of the journal of submission.

## EVIDENCE-BASED SURGERY

Surgical practice has been considered an art: ask 50 surgeons how to manage a patient and you will probably get 50 different answers. There is so much clinical information available that no surgeon can know it all. Evidence-based surgery is a move to find the best ways of managing patients using clinical evidence from collected studies. It was estimated that sufficient evidence to justify routine myocardial thrombolysis for heart attacks was available years before the randomised clinical studies that finally made it clinically acceptable; no one had gathered all the available information together.

Centres such as the Cochrane Collaboration have been collecting randomised trials and reviews to provide up-to-date information for clinicians. The Cochrane Library presently includes a database of systematic reviews, reviews of surgical effectiveness and a register of controlled trials. The *BJS* has been collecting surgical randomised trials on its website archive for 20 years (www.bjs.co.uk). As evidence accumulates, it is expected that this will gradually smooth out the differences between clinicians as the best way of managing patients becomes more obvious. Collecting published evidence together and analysing it often requires reviews of multiple randomised trials. These meta-analyses involve complex statistical analyses designed to interpret multiple findings and synthesise the results of multiple studies.

## FURTHER READING

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## ONLINE RESOURCES

AcoRD: https://www.gov.uk/government/publications/guidance-on-attributing-the-costs-of-health-and-social-care-research
CLAHRC: https://clahrcprojects.co.uk/about
Clinical Evidence: www.clinicalevidence.com
Cochrane Library: www.cochrane.org/index.htm
Concordat to Support Research Integrity: https://ukrio.org/revised-concordat-to-support-research-integrity-published
Consolidated Standards of Reporting Trials: http://www.consort-statement.org
Data Archive: http://www.data-archive.ac.uk
Eudract database: https://www.clinicaltrialsregister.eu
European Code of Conduct: https://allea.org/wp-content/uploads/2017/05/ALLEA-European-Code-of-Conduct-for-Research-Integrity-2017.pdf
GafREC: https://www.hra.nhs.uk/planning-and-improving-research/policies-standards-legislation/governance-arrangement-research-ethics-committees
Health Research Authority: http://www.hra.nhs.uk
Integrated Research Application System: https://www.hra.nhs.uk/about-us/committees-and-services/integrated-research-application-system
ISRCTN: http://www.isrctn.com
MHRA: https://www.gov.uk/government/organisations/medicines-and-healthcare-products-regulatory-agency
National Institute for Health and Care Excellence (NICE): https://www.nice.org.uk
NHS England audits: https://www.england.nhs.uk/clinaudit
REDCap: https://projectredcap.org
Scottish Intercollegiate Guideline Network (SIGN): www.sign.ac.uk
Singapore Statement on Research Integrity: https://wcrif.org/guidance/singapore-statement
Vascular Society: http://www.vascularsociety.org.uk